Chapter 2

Technical Soundness

2.1 Design overview

[**Project Name**] is a warning *chain*, not a sensor. It detects a rising wadi, decides how dangerous the situation is, delivers the warning through whichever channel still works, requires a responsible person to confirm it, escalates automatically when nobody does, and hands the evidence to the authority that decides on public alerts. Every choice traces back to the six requirements of Chapter 1: work without grid, internet or mobile network (R1); confirm receipt and act on silence (R2); give safe-action guidance (R3); limit false alarms (R4); be low-cost and locally repairable (R5); fit the institutional chain and never replace human decisions on mass alerts (R6).

Four principles follow. **Local first:** the danger signal at the crossing is produced and shown at the crossing, and no remote layer is needed for the beacon to fire. **Silence is never "all clear":** a missing message raises a fault and never lowers the risk level. **Two independent signals before alarming people:** one reading can only raise a watch. **Humans own decisions:** the system recommends to the regional committee and never issues a public alert itself.

Figure 2.1 shows the architecture and Figure 2.2 how it sits on the ground. Three solar nodes are placed along one wadi: **N1** upstream, **N2** midstream and **N3** at a road crossing. On the illustrative site (Oued Souhil, Nabeul) they sit at about 3.3, 4.9 and 6.15 km along the mapped channel, the last at the RR27 bridge [1]. They report by radio to a gateway on a public building near the RR27 crossing (exact site to be agreed with the municipality), which hosts the network server, the risk engine, the escalation engine, the event log, the alert exporter and a read-only dashboard.

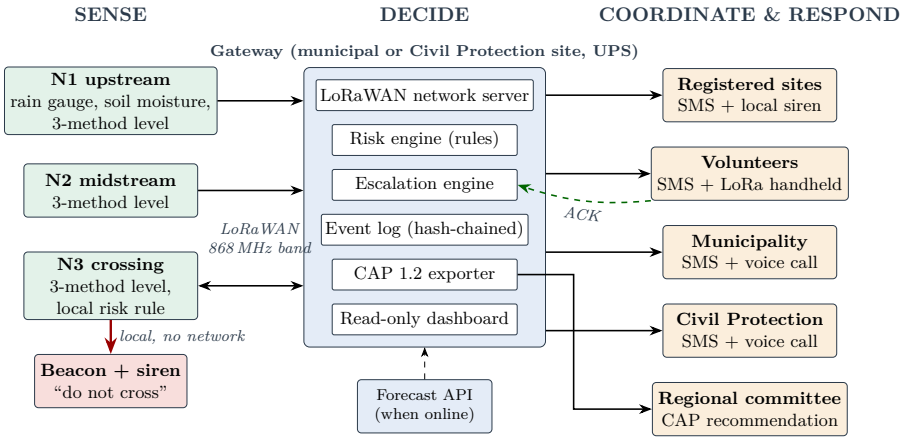


Figure 2.1: System architecture. N3 shows danger at the crossing even if the gateway is unreachable. The gateway adds the chain: rules across nodes, delivery, acknowledgement (dashed) and escalation, a tamper-evident log and a standard-format recommendation to the deciding authority.

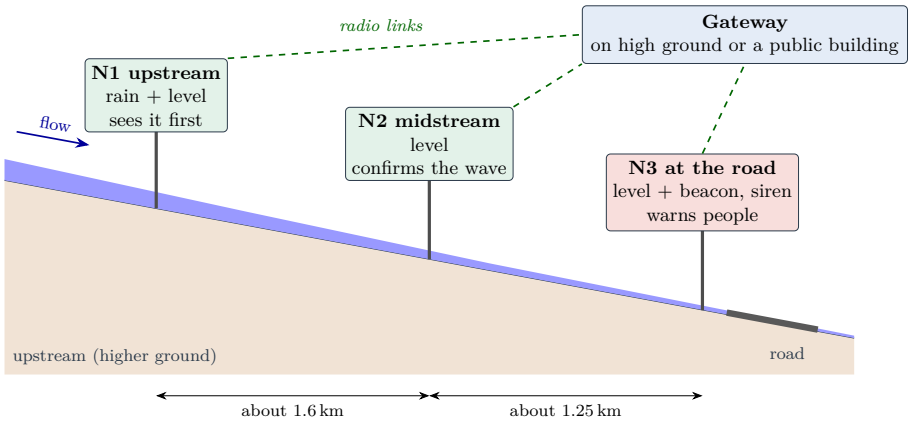


Figure 2.2: Side view of the wadi (not to scale). N1 sees the flood first, N2 confirms that the water is moving, and N3 warns people where they would meet it. Distances are from the illustrative site [1].

2.2 Innovation and efficiency

We do not claim a new sensor. Wadi level sensing over LoRa, SMS alerts, rate-of-rise monitoring and multi-level risk states are all published [2, 3, 4, 5], and acknowledgement with escalation exists in commercial, internet-based mass-notification platforms [6]. The closest system we found, a solar LoRa flood detector with online and offline modes, issues two severity tiers and was tested in the laboratory [5].

Our contribution is the **combination, designed for the conditions where warnings fail**: per-recipient acknowledgement with timed escalation from neighbourhood to Civil Protection; an acknowledgement path that needs no mobile network; agreement between independent signals before people are alarmed; fail-safe treatment of silence; and a Common Alerting Protocol (CAP) bridge into a human-decided national channel. In a search of literature and products between 22 September and 7 October 2026 we found no published low-cost flash-flood system combining these (Table 2.1).

Table 2.1: Comparison with the closest work found. ✓ = reported; – = not reported in the parts we read (absence of evidence, not evidence of absence).

Capability	[2]	[4]	[5]	[3]	[6]	**[Project Name]**
Level sensing over LoRa / LoRaWAN	✓	✓	✓	✓	–	✓
Rate of rise and multi-level risk states	–	✓	✓	–	–	✓
Local alarm without internet	–	–	✓	–	–	✓
Per-recipient acknowledgement, timed escalation	–	–	–	–	✓	✓
Acknowledgement without mobile network	–	–	–	–	–	✓
Independent-signal agreement before alarm	–	–	–	–	–	✓
CAP bridge to a human-decided channel	–	–	–	–	–	✓

Efficiency. Energy, airtime and money are spent only where they change outcomes: nodes sleep and sample faster only when risk rises; radio settings keep storm-mode reporting inside duty-cycle limits; SMS is reserved for a small registered group, since bulk SMS can be delayed [7] and public reach belongs to the national channel; and the rule-based logic runs on one single-board computer with no cloud dependency.

2.3 Sensing layer

Ultrasonic sensing alone can be disturbed by foam, turbulence and debris [8], and a node blinded during a storm is worse than no node. Each node therefore measures water level in three physically independent ways and votes (Table 2.2, Figure 2.3). The **ultrasonic sensor** hangs on an arm above the highest flood level and looks down: it times the echo of a sound pulse from the water surface and subtracts that distance from its mounting height. The **pressure sensor** sits at the bed inside a protective pipe: the deeper the water above it, the higher the pressure. The **float switches** are

fixed at three heights on a staff; each closes when the water reaches it, giving three yes-or-no answers that need no calibration.

Table 2.2: Sensing methods (manufacturer values, to be verified in our tests).

Method	Implementation	Weakness and mitigation
Ultrasonic level	Waterproof module on an arm (IP67, 3–450 cm, ≤8 mA [9])	Foam, spray, debris: median of 5 echoes, cross-check
Pressure level	Submersible 4–20 mA transmitter in a slotted steel pipe	Silt and impact: yearly cleaning
Float ladder	Three switches at Watch, Warning, Danger heights, supervised inputs	Three steps only: used as a hard check
Rain (N1)	Tipping bucket, counted in deep sleep	Clogging: mesh guard, monthly check
Soil moisture (N1, optional)	Capacitive probe, field-calibrated	Weakest signal: never triggers alone; removable

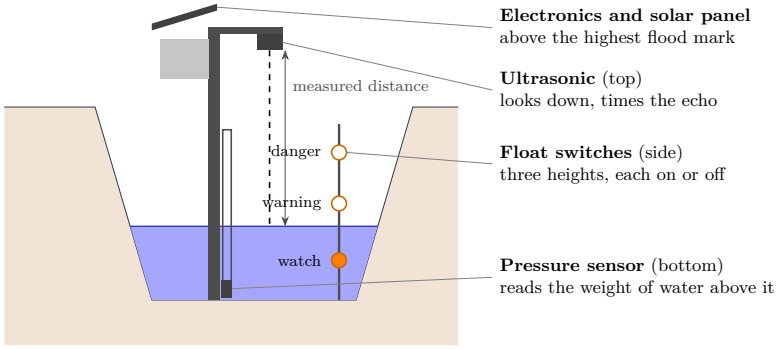


Figure 2.3: Cross-section of the channel at one node (not to scale): the three level-sensing methods are stacked at the same spot. Here the water has reached the Watch float only.

Fusion. The ultrasonic reading is the median of five echoes; the node cross-checks it with the pressure reading, and both are bounded by the float steps. If the two continuous sensors disagree for several samples by more than a tolerance that grows with depth (initially 30% of the depth or 10 cm, whichever is larger), the node raises a *degraded* flag and uses the *higher* of the two for alarms, never the lower. One rule decides Danger at a node: *two of the three methods above the danger height, counting only healthy methods.* If the other two are faulty or degraded, the Danger float alone is enough, so the node fails toward alarm. If the float is closed but both healthy continuous sensors agree that the water is well below it, the node raises Warning and a float-fault flag instead, so a float jammed by debris cannot alarm the town. A painted staff gauge at each node lets residents read the level themselves.

Soil moisture: an optional, deliberately weak signal. The probe at N1 is there for two reasons: the same rain produces more runoff on wet ground, so the system can react earlier; and in the dry season it records drought and water-resource data at almost no extra cost. It is also the weakest sensor in the design. One probe says little about a whole catchment, low-cost probes drift, and in semi-arid areas, including sites studied in Tunisia, runoff comes mainly from rain falling faster than the soil surface can absorb [10], so the rain gauge is likely the better predictor. We found no local data showing that it improves flood warning in Cap Bon. The design therefore gives it no authority: it can only adjust the rain thresholds, never raise an alert alone, and the system works unchanged without it. It is the first component to drop if cost or complexity must be reduced.

Electronics. Every node uses the same core, an ESP32-S3 microcontroller with a Semtech SX1262 LoRa transceiver (up to +22 dBm [11]), so firmware and spares are shared. A solar panel charges a lithium iron phosphate battery, chosen for its tolerance of heat inside a sunlit enclosure. Sensors are powered only while measuring; N3 adds a 12 V bus for the beacon and siren. Figure 2.4 shows the three interface circuits that matter most for reliability.

Firmware. At N1 and N2, sampling adapts to risk: every 10 minutes in Normal, every minute in Watch, every 10 seconds in Warning and Danger (initial values). N3, which protects the crossing,

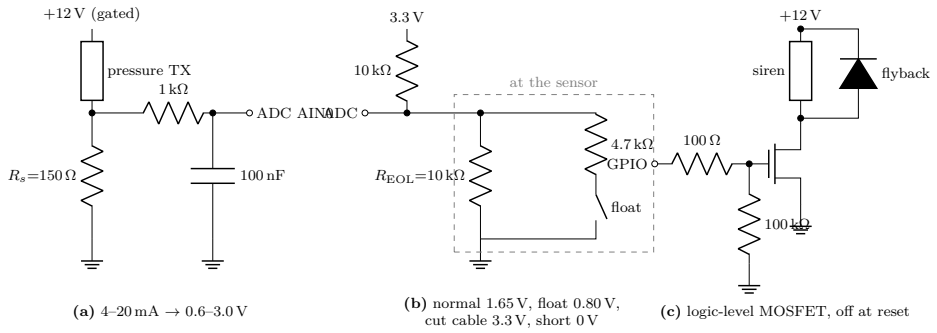


Figure 2.4: Interface circuits. (a) Pressure transmitter current read as a voltage by a 16-bit ADC. (b) Supervised float input: four distinct voltages, so a damaged cable reports a fault instead of "dry". (c) Beacon and siren driver, held off during reset.

never waits for the gateway to raise the risk level: it samples *every minute in all states* and every 10 seconds from Warning, so a storm that hits only the lower catchment, with no upstream signal and no gateway, is still seen within a minute. On every node a closing float also wakes the processor at once: the ESP32-S3 low-power coprocessor reads the supervised float inputs every second while the main core sleeps, and a high rate of rain-gauge tips wakes N1 the same way. N1 and N2 send a heartbeat at least every 10 minutes in Normal and with every report from Watch; N3 sends one every minute. The gateway marks a node *faulty* after three missed heartbeats, so a silent N3 is noticed within 3 minutes. N3 evaluates its own danger rule on every sample and drives the beacon and siren directly; it also listens continuously (LoRaWAN Class C), so the gateway can switch the beacon on early when N1 and N2 see the flood coming. Each node runs a hardware watchdog that resets hung firmware within seconds.

2.4 Communication layer

Network. Nodes use LoRaWAN, which gives per-device AES-128 keys, integrity codes and replay protection. A self-hosted ChirpStack server runs on the gateway, so the network works without internet [12]; a second gateway can be added for path diversity, as the server merges copies of the same message [13]. Each report is 16 bytes of data, 29 bytes on air.

Duty cycle. In the European rules commonly applied to this band, 868.0–868.6 MHz allows 25 mW at a 1% duty cycle and 869.4–869.65 MHz allows 500 mW at 10% [14]. Tunisian conditions are not yet confirmed: community tables conflict and the regulator's page is silent on this band [15, 16], so we are seeking confirmation and keep the band configurable. Table 2.3 shows the constraint: at SF10 a node could report only every 41 s. Storm-mode reports are therefore fixed at SF7–SF8 every 30 s, with state changes sent immediately as confirmed messages.

Table 2.3: Radio budget for a 29-byte frame at 125 kHz (airtime from the Semtech formula [17]; sensitivity from the noise-floor estimate with a 6 dB noise figure; margin over free-space loss on the longest node-to-gateway path, budgeted at 3 km, for +14 dBm, 2 and 5 dBi antennas, 2 dB cable loss).

	Airtime	Min. interval (1%)	Sensitivity	Max. path loss	Margin at 3 km
SF7	66.8 ms	6.7 s	−124.5 dBm	143.5 dB	43 dB
SF10	411.6 ms	41.2 s	−132.0 dBm	151.0 dB	50 dB
SF12	1646.6 ms	164.7 s	−137.0 dBm	156.0 dB	55 dB

Link feasibility. Nodes do not talk to each other; each one talks to the gateway, so the budget is set by the node-to-gateway paths, not by the spacing between nodes. With the gateway near the RR27 crossing, the straight-line distances on the mapped site are about 2.5 km for N1, 1.2 km for N2 and under 0.5 km for N3 [1]. We budget for 3 km to leave room for the exact gateway site: free-space loss is then about 101 dB, leaving 43–55 dB for terrain and vegetation. A comparable SX1262 network measured a 2.47 km link at −110 dBm [18], and LoRa path loss has been measured at 868 MHz in Tunisian oases [19]. Free-space margin is not enough on its own, because the nodes

sit on a valley floor and may have no line of sight to the gateway. Before installation, each path is therefore checked with a terrain profile (Radio Mobile on open elevation data) and then walk-tested. A path without line of sight gets a higher antenna mast at the node, a higher spreading factor (SF10 still allows one report a minute), or the optional second gateway on high ground upstream.

Graceful degradation. If the *internet* fails, everything local continues: rules on sensors only, SMS, voice calls, beacon and log. If the *mobile network* fails, the beacon and siren still fire, and volunteers still receive and acknowledge alerts on Meshtastic handhelds [20], an open-source LoRa mesh encrypted with AES-256 [21]; officials are alerted when the network returns, flagged as delayed. If the *gateway* fails, N3 notices that its confirmed messages are no longer acknowledged, keeps firing from its own rule, and shows a "gateway lost" status light for technicians; the maintenance team is told by the missing daily status message (Section 2.7). If *mains power* fails, nodes are solar and the gateway runs on its battery. If a *node goes silent*, the crossing is reported as "unmonitored", never as "safe".

2.5 Risk engine

The risk engine is a deterministic, rule-based state machine with four states (Figure 2.6). Signals form three *independent* groups: **M**, meteorological (a forecast flag from an open weather API [22] when online, or rain intensity at N1); **U**, upstream (level or rate of rise at N1 or N2); **D**, downstream (level or rate of rise at N3). One group can raise a Watch; a Warning needs two groups to agree; Danger needs two of the three healthy methods at N3 above the danger height (with the float fallback described under Fusion), or a danger level upstream with N3 rising. A restart never lowers the state (Section 2.7). The danger height is set from the site survey, relative to the road surface. All thresholds and hold times are initial values kept in one version-controlled file.

Why rules, and why more than one threshold. The engine uses no artificial intelligence: it compares measurements with thresholds. We chose rules over a trained model because there is no local training data and because Civil Protection must be able to read, audit and explain every decision. A single true-or-false test on one sensor would be simpler still, but it fails in four ways. One sensor can be wrong (foam, debris, a splash), so alarms need agreement. A level threshold at the crossing reacts only when the water is already there, so upstream rain and rate of rise are used to gain time. Water hovering near a threshold would switch the alarm on and off (Figure 2.5), so the alarm clears only below a lower level held for a set time. And a plain test reads a dead sensor as "below threshold", that is, safe; our rules treat silence as a fault.

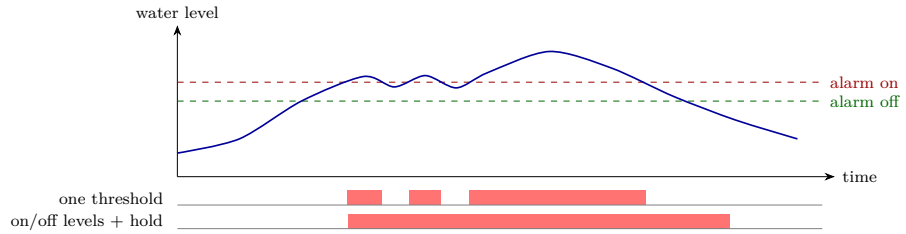


Figure 2.5: Why one threshold is not enough: the alarm (red bars) flickers three times while the water hovers around a single threshold; with separate on and off levels and a hold time, the same flood gives one continuous alarm.

Arrival estimate. The gateway records how long a rise takes to travel from N1 to N3 and, once calibrated on real events, can show an estimated arrival time as advice. We promise no lead time, which in small steep basins may be under an hour [23].

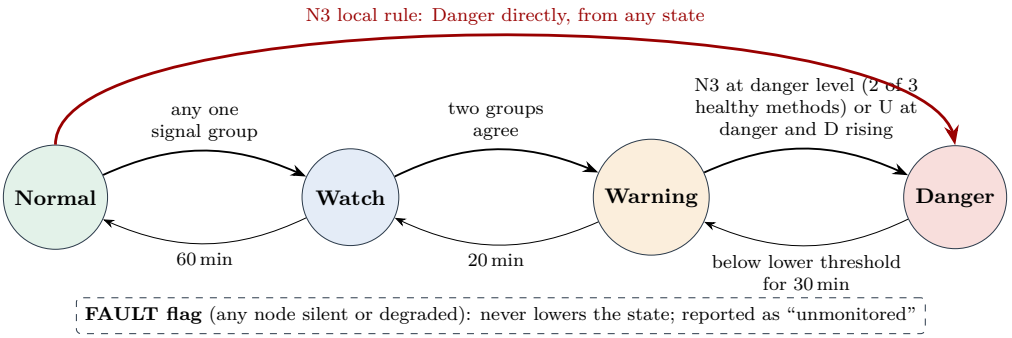


Figure 2.6: Risk state machine. Upgrades need more evidence as the state rises; downgrades need the signals to stay below a lower threshold for a hold time. Hold times are initial values.

2.6 Coordination: escalation and institution bridge

Each alert is a state machine with a timestamp on every transition (Table 2.4, Figure 2.7). Recipients answer `1` (received, acting) or `2` (cannot act); a `2` escalates at once. Volunteers answer by SMS or with the handheld button; officials are reached by SMS and, from tier T3, by an automated voice call answered with a key press. Only replies from registered numbers or paired handhelds count.

Table 2.4: Escalation ladder (timers are initial values to agree with Civil Protection).

Tier	Recipient	Triggered by	ACK	If no ACK
T0	Beacon and siren at N3	Danger (or N3 local rule)	–	–
T1	Registered sites	Danger	optional	–
T2	Zone volunteers (2–4)	Warning (information), Danger (action)	3 min	escalate to T3
T3	Municipal duty officer	T2 silent, or Danger lasting 10 min	3 min	escalate to T4
T4	Civil Protection, regional	T3 silent, or level still rising	3 min	repeat; log
T5	Regional committee	Danger: CAP recommendation, then updates	logged	–

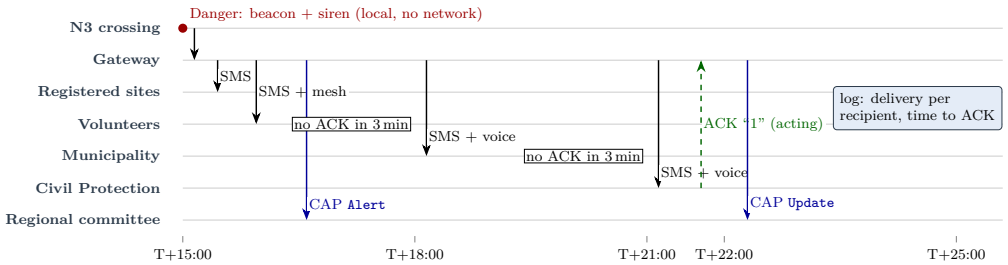


Figure 2.7: Escalation when nobody in the neighbourhood answers (illustrative times, not measured). The beacon fires first and needs no network; every message and acknowledgement is timestamped.

Message design. Messages carry a safe action, not just a level, because in 2018 a warning existed the day before and some residents still stayed on rooftops rather than cross floodwater [24]. Each message fits one SMS (160 Latin characters, or 70 in Arabic), names the place and says what to do, for example: *"DANGER Oued Souhil, RR27 bridge: do not cross. Move away from the wadi bed. Reply 1 if received."* Arabic and French texts are validated with Civil Protection before use.

Institution bridge. On Danger the gateway produces a Common Alerting Protocol 1.2 message [25] carrying the evidence (levels, rain, acknowledgement status, node health) and sends it to the regional committee, which decides on any public alert through the national channel [26, 27]. **[Project Name]** never sends public alerts itself. The message is signed, as CAP 1.2 permits [25], so the committee can verify its origin; demonstration messages use status `Exercise`.

Event log and dashboard. Every reading, state change, message and acknowledgement is appended to a local log in which each record contains the hash of the previous one, so later edits are detectable. The dashboard is a read-only view of this log; the system runs without it.

2.7 Power and resilience

The largest load is not the radio but waking the sensors every 10 s in storm mode. With our design assumptions (50 μA asleep, 100 mA for 1.5 s per sample, 120 mA while transmitting), N1 and N2 use about 9 mAh a day in Normal mode and 364 mAh in storm mode, so a 6 Ah battery gives about 13 storm days with no sun. The worst month at the site averages 4.5 kWh/m^2 per day on a tilted panel [28], so a 5 W panel returns about 4 Ah a day, roughly ten times the storm-mode need. N3 carries a 12 V, 20 Ah battery and a 30 W panel for the beacon and siren; sampling every minute and listening continuously add only about 0.2 Ah a day. Its processor has its own regulated supply with brown-out detection, so the current drawn by the siren cannot reset it. The gateway (about 10 W) runs on mains with a 100 Ah battery, about four days without power. All consumption figures will be replaced by measurements. Nodes are mounted above the highest recorded flood mark, in IP67 enclosures with hidden fixings and a tamper switch.

Restart during an event. A restart must neither lose an alert nor start it again from the beginning. The gateway writes every risk-state change and every alert (current tier, who was contacted, who acknowledged, next deadline) to its SQLite store *before* acting on it, and a service supervisor restarts any crashed service within seconds. At boot it restores this state: the risk level is the higher of the saved level and the newly computed one, and it can only go down through the normal hold times; each open alert resumes at the tier it had reached; timers that expired during the outage fire at once; recipients who had not yet answered receive one copy marked "resent after system restart", and those who had answered are not alerted again. The restart itself is logged and reported to the duty officer. At N3, the Danger state is also kept in memory that survives a reset. Outputs are off during reset (Figure 2.4c), but the firmware reads the sensors and the saved state within about a second of starting, and switches the beacon and siren back on if Danger still holds.

Health and maintenance alerts. Equipment problems go to a separate recipient, the municipal maintenance technician, so that they never mix with flood alerts. Each heartbeat carries the node's battery voltage and sensor status. The technician receives an SMS when a node's battery falls below 30%, a node goes silent or a sensor is degraded, mains power is lost at the gateway (with the estimated hours of battery left), and the gateway battery falls below 50% and 25%. At 25% the duty officer and Civil Protection are also told that the gateway will stop and that the crossing will then be protected only by N3's local rule. Before its battery runs out, the gateway saves its state, sends a last message and shuts down cleanly. A dead gateway cannot report its own failure, so it also sends the technician a short daily status message: if that message does not arrive, the gateway is down and is checked the same day.

2.8 Software, tools and methodology

Tools. Hardware is designed in KiCad and FreeCAD, with PVGIS for solar sizing. Firmware is built with PlatformIO and the RadioLib LoRaWAN stack. The gateway runs ChirpStack, an MQTT broker and our Python services: the risk and escalation engines as pure state machines, an SQLite event log, a CAP generator validated against the OASIS schema, and SMS and voice through the modem. The dashboard uses FastAPI and Leaflet with offline maps. Everything is open source and published on GitHub with continuous integration.

Methodology. We follow a light systems-engineering V-model: requirements R1–R6 are refined into system requirements, each with a design element and a verification test (Table 2.5). A failure-modes analysis of every component fed the degradation behaviour and the risk register (Chapter 4). Work runs in one-week sprints with a demo each week. The two state machines are tested before any hardware exists: property tests check, for example, that missing data can never lower the risk state, and a replay tool feeds recorded or synthetic flood curves into real nodes, so storms can be repeated on the bench.

2.9 Verification and feasibility

Table 2.5: Verification plan for Phase 2. Pass criteria are our targets.

#	Test	Pass criterion (target)	Req.
V1	Level accuracy against a ruler; foam and debris	Within 2 cm over 0–1 m; degraded flag, no false Danger	R4
V2	Terrain profile, then range walk test on each node-to-gateway path	≥99% delivery with retries	R1
V3	Power in each sampling profile	Within 20% of the budget	R1, R5
V4	Rain-only and splash scenarios (replay)	No Warning from a single signal group	R4
V5	Unanswered alert	T3 reached within 3 min ±5 s; every hop logged	R2
V6	Silent node, cut float cable, flat battery	Fault reported within 3 heartbeats	R1, R4
V7	Internet off, then mobile network off	Beacon fires; mesh ACK accepted; log intact	R1, R2
V8	CAP output; message comprehension; repair drill	Schema-valid and signed; safe action restated; part swapped with the manual alone	R3, R5, R6
V9	Storm at N3 only: N1 and N2 quiet, gateway off	Beacon on within 5 s of the Danger float closing	R1
V10	Danger rule: ultrasonic and pressure disabled with Danger float closed; then float forced closed with both sensors reading low	Danger in the first case; Warning and float fault in the second	R4
V11	Gateway power-cycled during escalation; N3 reset during Danger	Alert resumes at the same tier, nobody who answered is re-alerted, state not lowered; beacon back on within 2 s	R1, R2
V12	Mains off, node battery low, gateway switched off	Technician alerted within 5 min; missing daily status detected	R1, R5

Feasibility. Every component is off the shelf, and every software layer except our two state machines is a maintained open-source project. The hardware is deliberately simple (one core board for all nodes, common sensors, standard interfaces), which suits a team whose strength is software. The main risks are known and bounded: the radio band's legal status (configurable band, regulator consulted), sensor fouling (three methods, field tests), short lead time in small basins (local beacon, no lead-time promise) and coverage (terrain prediction, walk tests, second gateway). The Phase 2 prototype reproduces the full chain on a tray "wadi": three nodes and the gateway, a real radio link, the beacon and siren, real SMS and mesh acknowledgements, an unanswered alert escalating, a signed CAP file, and then the offline test V7.

2.10 Ethics in the technology

The commitments of Chapter 3 are built into the design. **Human authority:** the system recommends and the committee decides [26]. **Fail-safe:** silence raises a fault, outputs are off at reset, and a failed sensor reports "unmonitored", not "safe". **False alarms:** two independent signal groups and hold times protect trust, and false alarms are counted and published, because each one weakens the next warning [29]. **Privacy by design:** the household registry holds a phone number, a site, a language and an optional assistance flag, stored only on the gateway with consent under Tunisia's data-protection law [30]; the demonstration uses synthetic numbers. **Security:** encrypted and authenticated radio links, signed firmware and alerts, an allow-list for acknowledgements and a tamper-evident log. **Explainability:** every alert carries the readings that caused it, and the rules are public.

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